2.56 Forces: Connected Particles on Inclined Planes

Worked Solutions

1 a)

On A: $R(\nearrow)$: $T - 1.5 \times 9.8 \sin 30 = 1.5a$

On B: $R(\downarrow)$: $3.9 \times 9.8 - T = 3.9a$

$T = 15.925$ N, $a = \frac{343}{60} = 5.72$ ms^{-2} (3 SF)

$T = 15$ N, $a = 5.7$ ms^{-2} (2 SF – AQA)

Particle A is moving up the slope.

1 b)

On A: $R(\nearrow)$: $T - 2.1 \times 9.8 \sin 60 = 2.1a$

On B: $R(\downarrow)$: $0.5 \times 9.8 - T = 0.5a$

$T = 7.39$ N, $a = -4.97$ ms^{-2} (3 SF)

$T = 7.4$ N, $a = -5.0$ ms^{-2} (2 SF – AQA)

Particle A is moving down the slope.

1 c)

On A: $R(\nearrow)$: $T - 2 \times 9.8 \sin 40 = 2a$

On B: $R(\downarrow)$: $2 \times 9.8 - T = 2a$

$T = 16.1$ N, $a = 1.75$ ms^{-2} (3 SF)

$T = 16$ N, $a = 1.8$ ms^{-2} (2 SF – AQA)

Particle A is moving up the slope.

1 d)

On A: $R(\nearrow)$: $T - 40 \times 9.8 \sin 30 = 40a$

On B: $R(\downarrow)$: $20 \times 9.8 - T = 20a$

$T = 196$ N, $a = 0$ ms^{-2}

$T = 200$ N, $a = 0$ ms^{-2} (2 SF – AQA)

Particle A is not moving in either direction since a is zero.

2 a)

On B: $R(\downarrow)$: $5 \times 9.8 - T = 5a$

On A: $R(\nwarrow)$: $R - 2 \times 9.8 \cos 30 = 0$

$\Rightarrow R = \frac{49}{5}\sqrt{3} \Rightarrow F = 0.85 \times \frac{49}{5}\sqrt{3}$

On A: $R(\nearrow)$: $T - \frac{833}{100}\sqrt{3} - 2 \times 9.8 \sin 30 = 2a$

$T = 31.3$ N, $a = 3.35$ ms^{-2} (3 SF)

$T = 31$ N, $a = 3.4$ ms^{-2} (2 SF – AQA)

Particle A is moving up the slope.

2 b)

On A: $R(\nwarrow)$: $R - 7 \times 9.8 \cos 40 = 0$

$\Rightarrow R = 68.6 \cos 40$

$F = 0.7 \times 68.6 \cos 40$

On A: $R(\nearrow)$: $T - 48.02 \cos 40 - 7 \times 9.8 \sin 40 = 7 \times 0.4$

$\Rightarrow T = 83.7$ N (3 SF)

$T = 84$ N (2 SF – AQA)

On B: $R(\downarrow)$: $m \times 9.8 - 83.680\ldots = m \times 0.4$

$\Rightarrow m = 8.90$ kg (3 SF)

$m = 8.9$ kg (2 SF – AQA)

2 c)

On B: $R(\downarrow)$: $3 \times 9.8 - T = 3 \times -0.1$

$\Rightarrow T = 29.7$ N

On A: $R(\swarrow)$: $-29.7 - F + 7 \times 9.8 \sin 60 = 7 \times 0.1$

$\Rightarrow F = 29.0093427\ldots$

On A: $R(\nwarrow)$: $R - 7 \times 9.8 \cos 60 = 0 \Rightarrow R = 34.3$

$F = \mu R \Rightarrow \mu = 0.846$ (3 SF)

$\mu = 0.85$ (2 SF – AQA)

2 d)

On A: $R(\nwarrow)$: $R - 2k \times 9.8 \cos 30 = 0$

$\Rightarrow R = 9.8\sqrt{3}k$

$F = 0.5 \times 9.8\sqrt{3}k = 4.9\sqrt{3}k$

On A: $R(\nearrow)$: $T - 4.9\sqrt{3}k - 2k \times 9.8 \sin 30 = 2ka$

$\Rightarrow T - 2ka = (9.8 + 4.9\sqrt{3})k$

On B: $R(\downarrow)$: $9.8k - T = ka$

$T - 2ka + 9.8k - T = (9.8 + 4.9\sqrt{3})k + ka$

$-2ka + 9.8k = (9.8 + 4.9\sqrt{3})k + ka$

$9.8 - 9.8 - 4.9\sqrt{3} = 3a$

$a = -\frac{49}{30}\sqrt{3}$ ms^{-2}

$a = 2.83$ ms^{-2} (decelerating up the slope) (3 SF)

$a = 2.8$ ms^{-2} (decelerating up the slope) (2 SF – AQA)

3 a)

$\sin\theta = \frac{4}{5}, \cos\theta = \frac{3}{5}$

On B: $R(\uparrow)$: $T - 3 \times 9.8 = 3 \times 0.8 \Rightarrow T = 31.8$

On A: $R(\nwarrow)$: $R - 5 \times 9.8 \times \frac{3}{5} = 0 \Rightarrow R = 29.4$

On A: $R(\swarrow)$: $-31.8 - F + 5 \times 9.8 \times \frac{4}{5} = 5 \times 0.8$

$\Rightarrow F = 3.4$

$3.4 = 29.4\mu \Rightarrow \mu = 0.116$ (3 SF)

$\mu = 0.12$ (2 SF – AQA)

3 b)

$s = 0 + \frac{1}{2} \times 0.8 \times 5^2 = 10$ m

This assumes that box B does not reach the pulley, and that the boxes may be modelled as particles.

3 c)

Light means the string has zero mass.

Inextensible means that the acceleration is constant throughout.

4 a)

$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

On A: $R(\nwarrow\!\searrow)$: $R - m \times 9.8 \times \frac{4}{5} = 0 \Rightarrow R = 7.84m$

$F = 0.4 \times 7.84m = 3.136m$

On A: $R(\nearrow)$: $T - 3.136m - m \times 9.8 \times \frac{3}{5} = ma$

On B: $R(\downarrow)$: $2m \times 9.8 - T = 2ma$

$-3.136m - 5.88m + 19.6m = ma + 2ma$

$\Rightarrow a = 3.528$

$T = 2m \times 9.8 - 2m \times 3.528 = 12.544m$

4 b)

$1.5 = 0 + \frac{1}{2} \times 3.528 \times t^2 \Rightarrow t = 0.922$ seconds (3 SF)

$t = 0.92$ (2 SF – AQA)

4 c)

Phase 1: $t = 0.922$

$v^2 = 0^2 + 2 \times 3.528 \times 1.5 \Rightarrow v = \sqrt{10.584}$

Phase 2:

The tension is removed as the string will no longer be taut. The box goes up the slope before reaching a greatest height

On A: $R(\nearrow)$: $-3.136m - m \times 9.8 \times \frac{3}{5} = ma$

$\Rightarrow a = -9.016$

$0 = \sqrt{10.584} - 9.016t \Rightarrow t = 0.3608369577\ldots$

So total time:

$0.922\ldots + 0.360\ldots = 1.28$ seconds (3 SF)

1.3 seconds (2 SF – AQA)

5 a)

On A: $R(\uparrow)$: $R - 10 \times 10 = 0 \Rightarrow R = 100$

$F = 0.2 \times 100 = 20$

On A: $R(\rightarrow)$: $T - 20 = 10a$

On B: $R(\searrow)$: $3 \times 10 \times \sin 30 - T = 3a$

$15 - 20 = 13a \Rightarrow a = -\frac{5}{13} = -0.385$ ms^{-2} (3 SF)

$a = -0.3$ ms^{-2} (1 SF – AQA)

$T = \frac{210}{13} = 16.2$ N (3 SF)

$T = 20$ N (1 SF – AQA)

5 b)

On B: $R(\searrow)$: $4 \times 10 \sin 30 - T = 4 \times 1.2$

$\Rightarrow T = 15.2$ N

$T = 20$ N (1 SF – AQA)

On A: $R(\uparrow)$: $R - 5 \times 10 = 0 \Rightarrow R = 50$

On A: $R(\rightarrow)$: $15.2 - F = 5 \times 1.2 \Rightarrow F = 9.2$

$9.2 = \mu \times 50 \Rightarrow \mu = 0.184$

$\mu = 0.2$ (1 SF – AQA)

5 c)

On B: $R(\searrow)$: $2 \times 10 \sin 30 - 3 = 2a \Rightarrow a = 3.5$ ms^{-2}

On A: $R(\uparrow)$: $R - 0.5 \times 10 = 0 \Rightarrow R = 5$

On A: $R(\rightarrow)$: $3 - F = 0.5 \times 3.5 \Rightarrow F = 1.25$

$1.25 = \mu \times 5 \Rightarrow \mu = 0.25$

$\mu = 0.3$ (1 SF – AQA)

5 d)

On B: $R(\searrow)$: $10m \sin 30 - 12 = 2m$

$m = 4$ kg

On A: $R(\uparrow)$: $R - 10M = 0 \Rightarrow R = 10M$

$F = 0.4 \times 10M = 4M$

On A: $R(\rightarrow)$: $12 - 4M = 2M \Rightarrow M = 2$ kg

6 a)

$\sin\theta = \frac{4}{5}, \cos\theta = \frac{3}{5}$

On B: $R(\nearrow)$: $R_B - 8 \times 10 \times \frac{3}{5} = 0 \Rightarrow R_B = 48$

$F = 0.4 \times 48 = 19.2$

On B: $R(\searrow)$: $8 \times 10 \times \frac{4}{5} - T - 19.2 = 8a$

$44.8 - T = 8a$

On A: $R(\uparrow)$: $R_A - 10 \times 10 = 0 \Rightarrow R_A = 100$

$F = 0.2 \times 100 = 20$

On A: $R(\rightarrow)$: $T - 20 = 10a$

$44.8 - 20 = 8a + 10a \Rightarrow a = \frac{62}{45} = 1.38$ ms^{-2} (3 SF)

$a = 1$ ms^{-2} (1 SF – AQA)

6 b)

$T = \frac{304}{9} = 33.8$ N (3 SF)

$T = 30$ N (1 SF – AQA)

7 a)

On A: $R(\nwarrow)$: $R - 8 \times 10\cos 30 = 0 \Rightarrow R = 40\sqrt{3}$

$F = 0.05 \times 40\sqrt{3} = 2\sqrt{3}$

On A: $R(\swarrow)$: $-T - 2\sqrt{3} + 8 \times 10\sin 30 = 8a$

On B: $R(\nwarrow)$: $T - 3 \times 10\sin 45 = 3a$

$-2\sqrt{3} + 8 \times 10\sin 30 - 3 \times 10\sin 45 = 11a$

$a = 1.39$ ms^{-2} (3 SF)

$a = 1$ ms^{-2} (1 SF – AQA)

7 b)

Block A is going down the slope and block B is going up the slope.

7 c)

$T = 25.4$ N (3 SF)

$T = 30$ N (1 SF – AQA)

7 d)

On A: $R(\swarrow)$: $-T - 2\sqrt{3} + 8 \times 10\sin 30 = 0$

$\Rightarrow T = 40 - 2\sqrt{3}$

On B: $R(\nwarrow)$: $40 - 2\sqrt{3} - m \times 10\sin 45 = 0$

$m = 5.166956301\ldots$

$m = 5.17$ kg (3 SF)

$m = 5$ kg (1 SF – AQA)

8 a)

$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

On A: $R(\nwarrow)$: $R_A - 2 \times 10 \times \frac{4}{5} = 0 \Rightarrow R_A = 16$

On A: $R(\swarrow)$: $-16\mu - T + 2 \times 10 \times \frac{3}{5} = 2 \times 0.1$

$T + 16\mu = 11.8$

$\sin\alpha = \frac{4}{5}, \cos\alpha = \frac{3}{5}$

On B: $R(\nearrow)$: $R_B - 0.5 \times 10 \times \frac{3}{5} = 0 \Rightarrow R_B = 3$

On B: $R(\nwarrow)$: $T - 3 \times \frac{1}{2}\mu - 0.5 \times 10 \times \frac{4}{5} = 0.5 \times 0.1$

$T - \frac{3}{2}\mu = 4.05$

$-16\mu - \frac{3}{2}\mu = -7.75 \Rightarrow \mu = \frac{31}{70} = 0.443$ (3 SF)

$\mu = 0.4$ (1 SF – AQA)

8 b)

$T = \frac{33}{7} = 4.71$ N (3 SF)

$T = 5$ N (1 SF – AQA)

9 a)

$\sin\theta = \frac{5}{13}, \cos\theta = \frac{12}{13}$

On B: $R(\uparrow)$: $T - 9.8 \times 2 = 2 \times 0.4 \Rightarrow T = 20.4$ N

$T = 20$ N (2 SF – AQA)

9 b)

On A: $R(\nwarrow)$: $R - 10 \times 9.8 \times \frac{12}{13} + 30 \times \frac{5}{13} = 0$

$\Rightarrow R = \frac{1026}{13}$

On A: $R(\swarrow)$: $30 \times \frac{12}{13} - F + 10 \times 9.8 \times \frac{5}{13} - 20.4 = 10 \times 0.4$

$\frac{360}{13} - F + \frac{490}{13} - 20.4 = 4$

$\Rightarrow F = \frac{2664}{65}$

$\frac{2664}{65} = \frac{1026}{13}\mu \Rightarrow \mu = \frac{148}{245} \approx 0.519$ as req.

9 c)

The second box is does not have any resistances acting upon it. The string which passes over the pulley is parallel to the slope.

10 a)

On A: $R(\nwarrow)$: $R_A - 10 \times 9.8 \times \cos 30 = 0$

$\Rightarrow R_A = 49\sqrt{3}$

$F_A = 0.1 \times 49\sqrt{3} = 4.9\sqrt{3}$

On A: $R(\swarrow)$: $10 \times 9.8 \sin 30 - T_1 - 4.9\sqrt{3} = 10a$

On B: $R(\uparrow)$: $R_B - 5 \times 9.8 = 0 \Rightarrow R_A = 49$

$F_B = 0.4 \times 49 = 19.6$

On B: $R(\leftarrow)$: $T_1 - 19.6 - T_2 = 5a$

On C: $R(\nearrow)$: $R_C - 2 \times 9.8 \times \cos 45 = 0$

$\Rightarrow R_C = 9.8\sqrt{2}$

$F_C = 0.2 \times 9.8\sqrt{2} = 1.96\sqrt{2}$

On C: $R(\nwarrow)$: $T_2 - 2 \times 9.8 \sin 45 - 1.96\sqrt{2} = 2a$

$40.51295104\ldots - T_1 = 10a$

$T_1 - 19.6 - T_2 = 5a$

$T_2 - 16.63115149 = 2a$

$40.51295104\ldots - 19.6 - 16.63115149 = 17a$

$\Rightarrow a = 0.2518705618\ldots$

$a = 0.252$ ms^{-2} (3 SF)

$a = 0.25$ ms^{-2} (2 SF – AQA)

10 b)

$T_1 = 37.9942\ldots$

$T_1 = 38.0$ N (3 SF)

$T_1 = 38$ N (2 SF – AQA)

10 c)

$T_2 = 17.1349\ldots$

$T_2 = 17.1$ N (3 SF)

$T_2 = 17$ N (2 SF – AQA)

11 a)

$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

On A: $R(\nwarrow)$: $R - 2 \times 10 \times \frac{4}{5} + 20 \times \frac{3}{5} = 0 \Rightarrow R = 4$

$F = 0.5 \times 4 = 2$

On A: $R(\nearrow)$: $T - 2 \times 10 \times \frac{3}{5} - 2 - 20 \times \frac{4}{5} = 2a$

$T - 30 = 2a$

On B: $R(\uparrow)$: $4 \times 10 - T = 4a$

$a = \frac{5}{3}$ ms^{-2}

Acceleration is $\frac{5}{3}$ ms^{-2} up the slope

1.67 ms^{-2} (3 SF) up the slope

2 ms^{-2} (1 SF – AQA) up the slope

11 b)

$T = \frac{100}{3} = 33.3$ N (3 SF)

$T = 30$ N (1 SF – AQA)

11 c)

On B: $R(\downarrow)$: $1 \times 10 - R = 1 \times \frac{5}{3}$

$R = \frac{25}{3} = 8.33$ N $\therefore$ By N3L: 8.33 N

8 N (1 SF – AQA)

11 d)

The container would have mass. So the force exerted on the container by box C would decrease.

12 a)

$\sin\theta = \frac{5}{13}, \cos\theta = \frac{12}{13}$

On A: $R(\nwarrow)$: $R - kmg \times \frac{12}{13} = 0 \Rightarrow R = \frac{12}{13}kmg$

$F = \frac{1}{12} \times \frac{12}{13}kmg = \frac{1}{13}kmg$

On A: $R(\swarrow)$: $-T - \frac{1}{13}kmg + kmg \times \frac{5}{13} = kma$

On B: $R(\uparrow)$: $T - mg = ma$

$-mg - ma - \frac{1}{13}kmg + kmg \times \frac{5}{13} = kma$

$-g - a - \frac{1}{13}kg + \frac{5}{13}kg = ka$

$-g + \frac{4}{13}kg = ka + a$

$\left(\frac{4}{13}k - 1\right)g = a(k+1)$

$a = \frac{(\frac{4}{13}k-1)g}{k+1} = \frac{(4k-13)g}{13k+13}$ as req.

12 b)

The rope is modelled as light and inextensible, so that it has no mass and acceleration is constant through the system. The boxes are modelled as particles.

13 a)

$0.5 = 0 + \frac{1}{2} \times a \times 2^2 \Rightarrow a = \frac{1}{4}$ ms^{-2}

13 b)

$\sin\theta = \frac{7}{25}, \cos\theta = \frac{24}{25}$

On B: $R(\downarrow)$: $mg - T = m \times \frac{1}{4}$

$T = mg - \frac{1}{4}m = \left(g - \frac{1}{4}\right)m$ as req.

13 c)

On A: $R(\nwarrow)$: $R - 0.05mg \times \frac{24}{25} = 0 \Rightarrow R = 0.048mg$

On A: $R(\nearrow)$:

$\left(g - \frac{1}{4}\right)m - F - 0.05mg \times \frac{7}{25} = 0.05m \times \frac{1}{4}$

$F = mg - 0.25m - 0.014mg - 0.0125m$

$F = (0.986g - 0.2625)m$

$F = \mu R$

$(0.986g - 0.2625)m = \mu \times 0.048mg$

$\mu = \frac{0.986g-0.2625}{0.048g} = \frac{493}{24} - \frac{173}{32g}$

13 d)

The string is parallel to the slope.

14 a)

$\sin\theta = \frac{3}{5}$

On A: $R(\uparrow)$: $R_A - mg = 0 \Rightarrow R_A = mg$

$F = \mu mg$

On A: $R(\rightarrow)$: $T - \mu mg = ma$

$T = ma + \mu mg$

On B: $R(\searrow)$: $kmg \times \frac{3}{5} - T = kma$

$\frac{3}{5}kmg - ma - \mu mg = kma$

$\frac{3}{5}kg - a - \mu g = ka$

$\frac{3}{5}kg - \mu g = ka + a$

$\left(\frac{3}{5}k - \mu\right) g = (k+1)a$

$a = \frac{(\frac{3}{5}k-\mu)g}{k+1} = \frac{(3k-5\mu)g}{5k+5}$

14 b)

$\frac{3}{5}k - \mu > 0 \Rightarrow k > \frac{5}{3}\mu$

15 a)

$\mu_A = 3\mu$ and $\mu_B = \mu$

On A: $R(\nwarrow\!\searrow)$: $R_A - kmg\cos\theta = 0 \Rightarrow R_A = kmg\cos\theta$

On A: $R(\nearrow)$: $T - 3\mu kmg\cos\theta - kmg\sin\theta = 0$

On B: $R(\nearrow)$: $R - mg\cos\alpha = 0 \Rightarrow R_B = mg\cos\alpha$

On B: $R(\searrow)$: $-\mu mg\cos\alpha - T + mg\sin\alpha = 0$

$-\mu mg\cos\alpha - 3\mu kmg\cos\theta - kmg\sin\theta + mg\sin\alpha = 0$

$-\mu\cos\alpha - 3\mu k\cos\theta - k\sin\theta + \sin\alpha = 0$

$-\mu\cos\alpha + \sin\alpha = 3\mu k\cos\theta + k\sin\theta$

$\sin\alpha - \mu\cos\alpha = k(3\mu\cos\theta + \sin\theta)$

$k = \frac{\sin\alpha-\mu\cos\alpha}{\sin\theta+3\mu\cos\theta}$ as req.

15 b)

$1.1 = \frac{\sin 40-\mu\cos 40}{\sin 30+3\mu\cos 30}$

$\Rightarrow 1.1\sin 30 + 3.3\mu\cos 30 = \sin 40 - \mu\cos 40$

$\Rightarrow 3.3\mu\cos 30 + \mu\cos 40 = \sin 40 - 1.1\sin 30$

$\mu = \frac{\sin 40-1.1\sin 30}{3.3\cos 30+\cos 40} = 0.0256$ (3 SF)

16)

$a > 0$

On B: $R(\uparrow)$: $T - mg = ma \Rightarrow T - mg > 0 \Rightarrow T > mg$

On A: $R(\nwarrow)$: $R - 4mg\cos 30 + P\sin 30 = 0$

$\Rightarrow R = 2\sqrt{3}mg - \frac{1}{2}P$

$\Rightarrow F = 0.5 \times \left(2\sqrt{3}mg - \frac{1}{2}P\right) = \sqrt{3}mg - \frac{1}{4}P$

On A: $R(\swarrow)$:

$P\cos 30 - \sqrt{3}mg + \frac{1}{4}P - T + 4mg\sin 30 > 0$

$\frac{\sqrt{3}}{2}P - \sqrt{3}mg + \frac{1}{4}P - T + 2mg > 0$

$\left(\frac{\sqrt{3}}{2} + \frac{1}{4}\right)P + (2 - \sqrt{3})mg - T > 0$

$\left(\frac{\sqrt{3}}{2} + \frac{1}{4}\right)P > T - (2 - \sqrt{3})mg$

$(2\sqrt{3} + 1)P > 4T - (8 - 4\sqrt{3})mg$

$P > \frac{4T - (8 - 4\sqrt{3})mg}{2\sqrt{3}+1}$

$P > \frac{4mg - (8 - 4\sqrt{3})mg}{2\sqrt{3}+1}$

$P > \frac{(4\sqrt{3} - 4)mg}{2\sqrt{3}+1}$

$P > \frac{4}{11}(7 - 3\sqrt{3})mg$